

National University of Computer and Emerging Sciences



Assignment No. 5

Design and Analysis of Algorithms

Graphs & Spanning Trees

CS2009

Spring 2024

Deadline: May 10, 2024

Submission Instructions:

- All problems must be solved and submitted in hand-written format.
- Use A4 size papers with assignment front paper.
- This is an individual assignment.
- **Plagiarism is strictly prohibited.**
- **Please do not use any AI tool and do not copy from others.**
- **Just analyze the problem and then brainstorm the solution.**

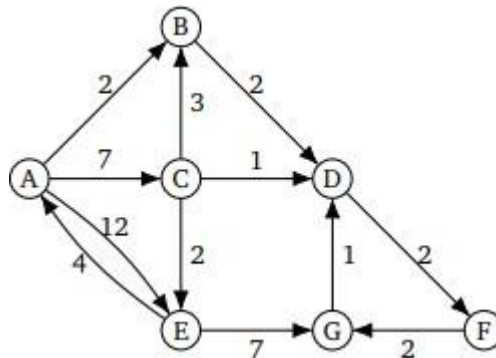
Give the following in Question 1-6

- Dry-run the examples with the algorithm
- Proper Steps of each algorithm you are applying
- Highlight the Final Graph or Result

Question 01:

(10 Marks)

Here is a graph. What are the final cost and resulting Spanning Tree if we run Prim's and Kruskal algorithm starting on node A? How many Spanning trees there can be?

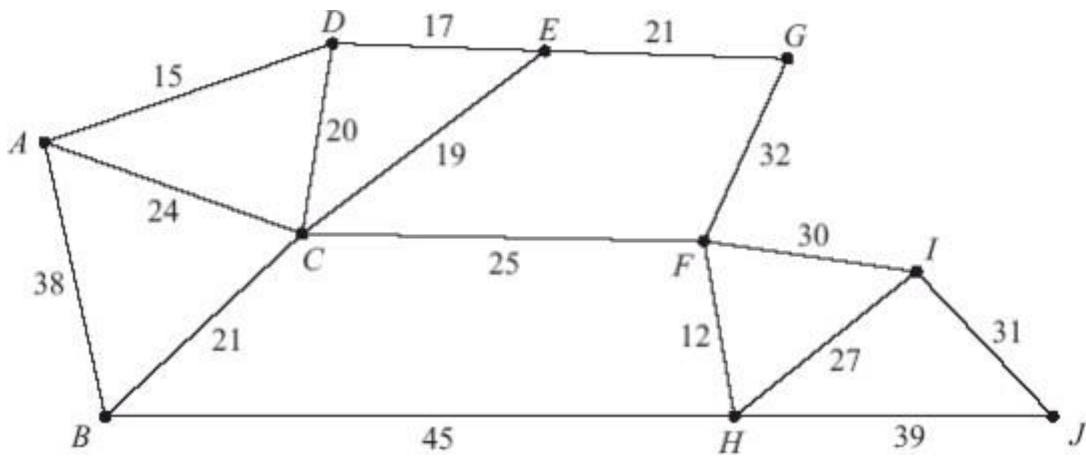


Question 02:

(10 Marks)

Find a minimum spanning tree for the network in the diagram below, showing clearly the order in which you selected the arcs for your tree, using

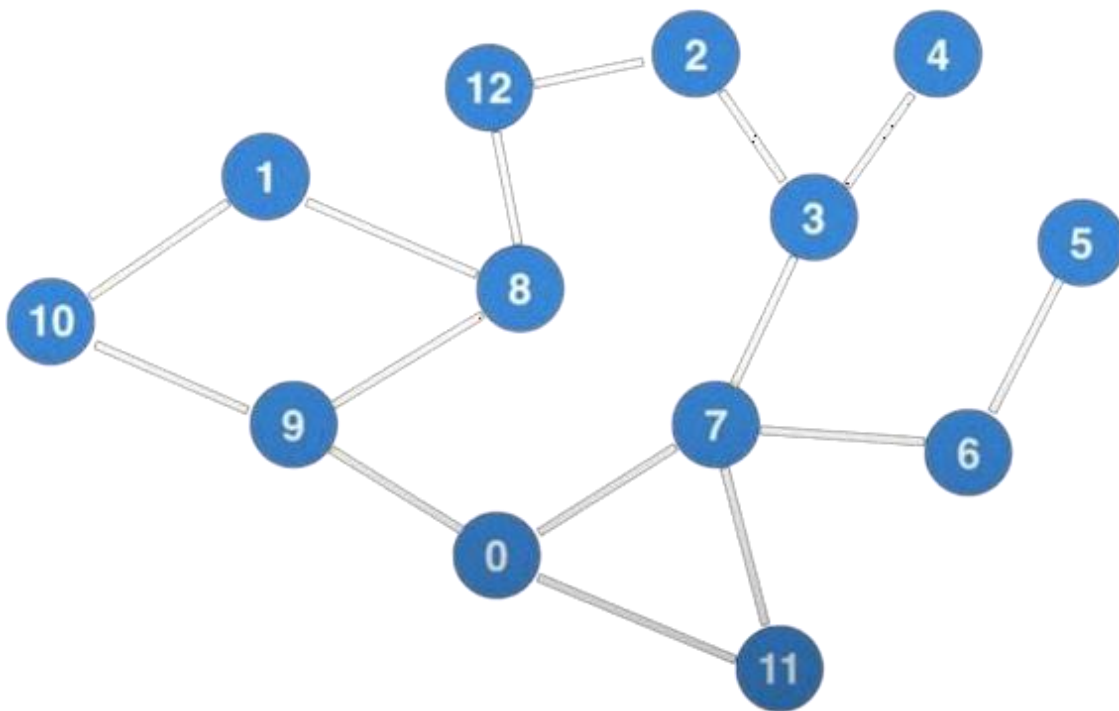
- Kruskal's algorithm
- Prim's algorithm, starting from H.



Question 03:

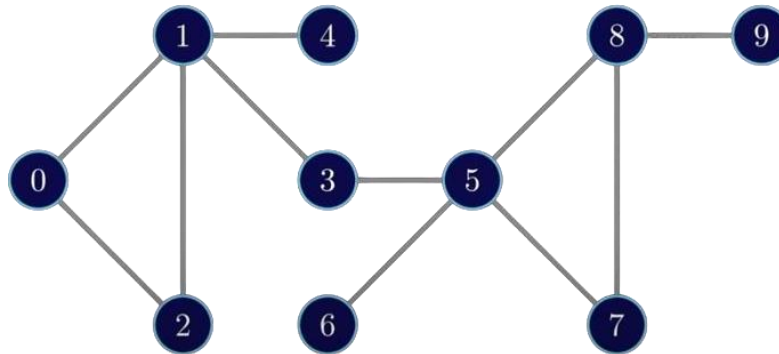
(10 Marks)

Run BFS and DFS on the below graph and show each and every step. Traverse to each node and show the steps. Assume the Starting node **8**



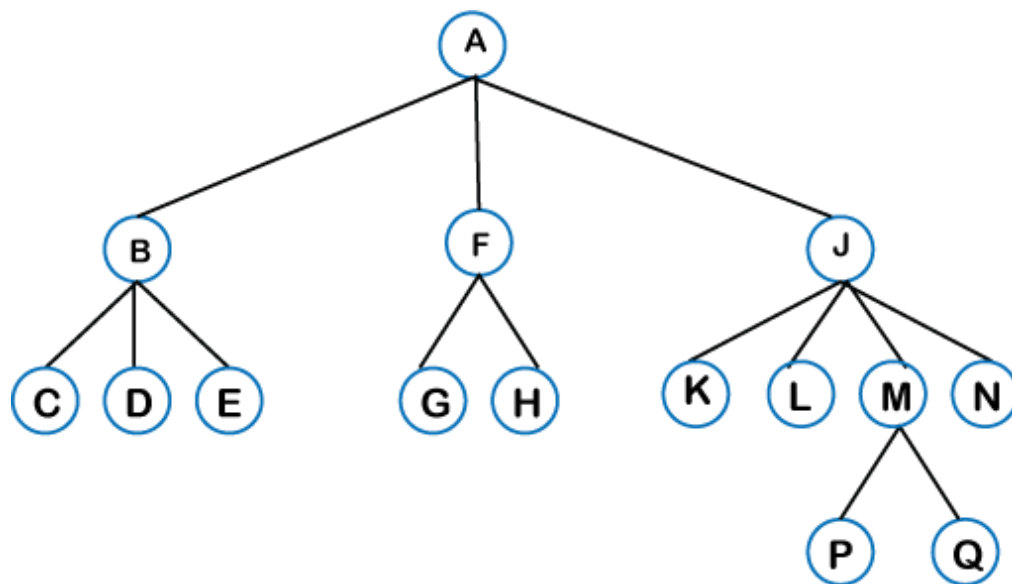
Question 04:**(10 Marks)**

Run BFS and DFS starting from the **Node 0** on the below graph and show each and every step. Traverse to each node and show the steps.

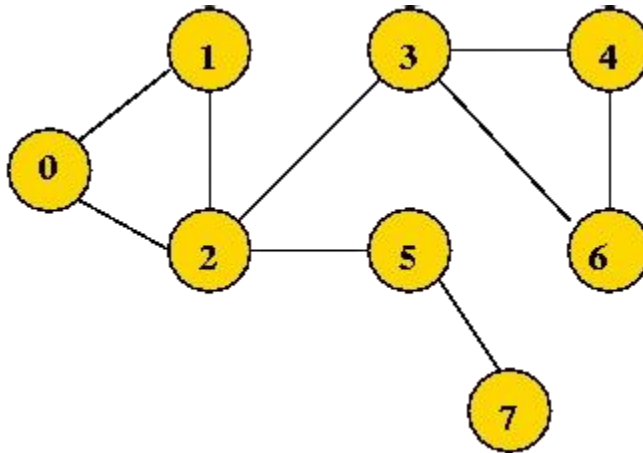
**Question 05:****(10 Marks)**

Find the following in the below graph:

- Degree
- Level
- Inner Nodes
- Leaf Nodes
- Define a path from node C to P
- Ancestors of M
- Descendants of J

**Question 06:****(10 Marks)**

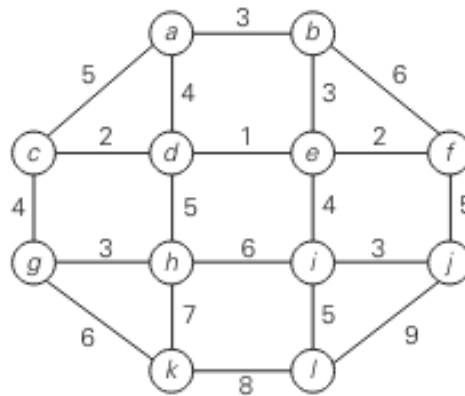
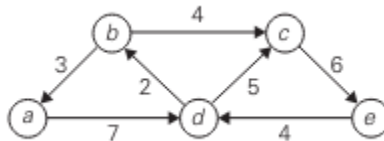
Detect the cycle in the below Undirected graph using BFS & DFS. Assume the starting node 5 (**if needed**).



Question 07:

(10 Marks)

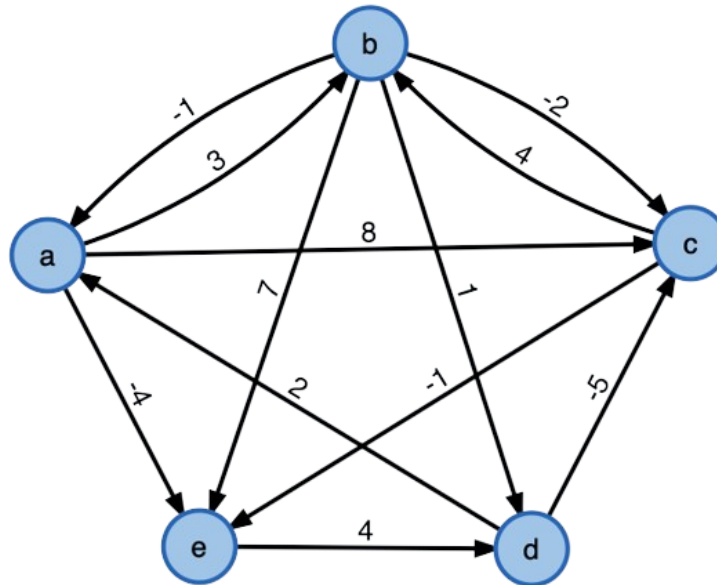
Solve the following instances of the single-source shortest-paths problem using Dijkstra Algorithm with vertex 'a' as the source



Question 08:

(10 Marks)

Solve the following instances of the single-source shortest-paths problem using both **Floyd Warshall** and **Bellman Ford** algorithm with vertex 'c' (if needed) as the source



Question 09:

(10 Marks)

Solve the following instances of the single-source shortest-paths problem using both **Floyd Warshall** and **Bellman Ford** algorithm with vertex 'A' (if needed) as the source.

